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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 10/10/2019 |
| **PI Name:** | Barstow, Thomas Ph.D. | | |
| **Project Title:** | Identification of Aerobic Fitness Standards for Exploration Mission Tasks | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Manhattan | **State:** | KS |
| **Zip Code:** | 66506-0109 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 02/24/2016 | **End Date:** | 09/30/2018 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 3 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 3 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
| **Contact Email:** | Peter.norsk@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 9/30/2018 per NSSC information (Ed., 3/26/18)<br>NOTE: Change in period of performance per NSSC funding information (original POP per Element was 12/8/2015-11/30/2017)--Ed., 4/12/16 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Ade, Carl Ph.D. ( Kansas State University ) | | |
| **Grant/Contract No.:** | NNX16AF66A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The physical demands of exploration mission tasks, like any job, can vary depending on the task and environment. During the original Apollo missions certain tasks like photography needed little effort whereas construction and shoveling type tasks required increased physical exertion. Planned missions to terrestrial surfaces (e.g., asteroid, the moon, and Mars), where crewmembers will spend an extended period of time in a microgravity setting, may result in a plausible scenario in which at some point within the mission critical tasks may become physically challenging enough that a decrease in performance and safety results. Therefore, it is critical to understand the relationship between the ability to complete specific mission critical tasks and aerobic exercise capacity. To mitigate the risk of an unacceptable scenario in which crewmembers are unable to perform critical mission tasks due to inadequate aerobic exercise capacity, a set of easily administered predictor tests must be developed and used to evaluate both countermeasure effectiveness and ultimately, whether a crewmember is ready and capable of performing various physiologically demanding tasks. Therefore, the overall goal of this proposed work is twofold: develop a thorough understanding of the physiological parameters associated with the ability to complete simulated mission critical tasks, and develop a battery of predictor tests that will provide critical information regarding a crewmember's readiness and capability to perform these tasks.<br>Specific Aims<br><br>1. Determine values and ranges for aerobic fitness variables that can individually or in aggregate (a) map to the level of success for a given mission critical task and (b) provide an aerobic fitness standard.<br><br>2. Using the information obtained from Specific Aim 1, design predictor testing protocols that can be performed with in-flight hardware and astronaut time constraints. |
| **Rationale for HRP Directed Research:** | This project is in direct response to both the August 2014 HSRB requirement and the baselined HRP Path to Risk Reduction milestone of providing updates to the NASA Fitness For Duty Standards. Specifically, the HSRB asked for an update to the aerobic capacity standard. The new standard should be tied to fitness for duty for exploration tasks and provide a quantitative index of readiness to perform key exploration tasks. This research to address this requirement will leverage upon our similar HRP-funded work "Standardized 'Pre-flight' Exercise Tests to Predict Performance during Extravehicular Activities in a Lunar Environment" (Barstow, NNX10AK60G). This new follow-on research requires specific expertise, technologies, and data analytics that have developed within the Thomas Barstow Laboratory at Kansas State University. Therefore the work reflects "focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal." Thus, the proposed work qualifies for directed research under the "Highly Constrained Research" category in the HRP Unique Processes, Criteria, and Guidelines (UPCG). |
| **Research Impact/Earth Benefits:** | The results of these studies will help identify which key components of physical fitness are required to perform different physical tasks. These results will, therefore, be applicable in a wide variety of settings, from rehabilitation to athlete evaluation, to determining the relative preparedness of astronauts for in-flight and destination extravehicular activity (EVA). These insights will be especially important when astronauts return to a gravitational environment, either on Earth or at their destination. These results will provide target information regarding minimum required strength and endurance from which in-flight and destination exercise countermeasures can be based. The strategy employed here can also function as a template for approaching the establishment of field tests for other occupations in which there is a demand for minimal physical performance, such as what has been done for firefighters and police officers. |
| **Task Progress:** | The proposed projects have been completed and data published. The general findings for each project are as follows: Project 1: Prediction of planetary mission task performance for long-duration spaceflight. Introduction: Long duration spaceflight elicits a cascade of physiological adaptations that creates a problematic state of "spaceflight deconditioning" that may limit an astronaut's ability to perform aerobically demanding tasks upon return to terrestrial surfaces (e.g., the moon, Mars, Earth). This study aimed to determine values and ranges for key aerobic fitness variables that can individually map the level of success for planetary mission tasks performance for long-duration spaceflight and use this information to develop a predictor-testing model that can be performed with in-flight equipment. Methods: We studied a group of 45 men and women who completed a series of mission critical tasks; a surface traverse task and a hill climb task. Participants performed each mission task at a low and moderate intensity designed to elicit specific metabolic responses similar to what is expected for ambulation in Lunar and Martian gravities, respectively. Aerobic fitness was characterized via cycling and rowing VO2peak, ventilatory threshold (VT), and critical power. Logistic regression and receiver operating characteristic (ROC) curve analysis were used to determine the cutoff thresholds for each aerobic fitness parameter that accurately predicted task performance. Results: The low and moderate intensity traverse tests estimated a VO2 response of 23±1 ml kg-1 min-1 and 29±1 ml kg-1 min-1. VO2peak, VT, and critical power all showed high sensitivity and specificity for identifying individuals who could not complete the mission tasks. Therefore, minimum fitness thresholds for each task at each intensity were determined for each of the submaximal and maximal fitness variables. Conclusion: In summary, we identified aerobic fitness thresholds below which task performance was impaired for both low and moderate intensity mission critical tasks. Specifically, cycling VO2peak, VT, and rowing CP could each be used to predict task failure.<br><br>Project 2: Prediction of Emergency Capsule Egress Performance. Introduction: Critical mission tasks for Martian exploration have been identified and include specific duties that astronauts will have to perform despite any adverse effects of chronic microgravity. Specifically, astronauts may have to perform an emergency capsule egress upon return to Earth, which places specific demands on compromised cardiovascular and neuromuscular systems. Therefore, the purpose of this project was to determine the relationship between cardiorespiratory fitness factors and simulated capsule egress time. Methods: 15 participants volunteered for this study. VO2peak and peak power output (PPO) were determined on cycle and rowing ergometers. Critical power (CP) was determined by a 3 min all-out rowing test. Subjects then performed an emergency capsule egress on a mock-up of NASA's Orion space capsule. Peak metabolic data were compared between the cycling and rowing tests. Pearson's Correlation was used to identify relationships between egress time and VO2peak, PPO, and CP. Results: VO2peak, VCO2peak, and minute ventilation were not different between cycling and rowing tests. Cycling elicited a greater PPO than the rowing test. Egress time was negatively correlated to cycling and rowing VO2peak (r=-0.61) and cycling and rowing PPO (r=-0.56 and r=-0.74, respectively), but not CP. Conclusions: Our data suggest that rowing PPO/kg was a strong predictor of egress time. Although individuals with higher PPO/kg were able to finish the task in less time, even sedentary individuals with low fitness levels (VO2peak<20 ml/kg/min) were able to complete the egress within 2 mins. This suggests that if minimum fitness is maintained, completing the emergency egress should be successful. |

| Bibliography Type: | Description: (Last Updated: 01/23/2020) |
|---|---|
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